OAVS TGT COMPUTER TEACHER EXAMINATION

MCQ Question Bank — Programming Concepts of Basic Java

Total Questions: 159 (Volume 1)

Coverage: Identifiers & Keywords · Data Types · Coding Conventions · Expressions · Control Structures & Decision-Making · Arrays & Methods

Difficulty Mix: 25% Easy · 50% Moderate · 25% Difficult / Tricky

Format: Every question includes 4 options, the correct answer, and a short explanation

Strictly limited to the syllabus areas of Programming Concepts of Basic Java as specified. No advanced OOP, exception handling, collections, or other out-of-scope Java topics are included.

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Section 1 — Identifiers and Keywords

Q1 | Identifiers and Keywords | Definition

What is a Java identifier?

A. A type of Java operator

B. A built-in Java data type

C. A name given to variables, methods, classes, and other user-defined elements

D. A reserved word used by the Java compiler

Ans: C

Explanation: *An identifier is simply the name a programmer chooses for a variable, method, class, or other program element.*

Q2 | Identifiers and Keywords | Naming Rule

Which of the following is a valid Java identifier?

A. 2value

B. class

C. my-value

D. total_amount

Ans: D

Explanation: *total_amount begins with a letter, contains only letters and underscore, and is not a reserved keyword.*

Q3 | Identifiers and Keywords | Naming Rule

Which identifier is INVALID because it starts with a digit?

A. 10value

B. value10

C. value_10

D. _10value

Ans: A

Explanation: *Java identifiers cannot begin with a digit; they must start with a letter, underscore, or dollar sign.*

Q4 | Identifiers and Keywords | Keyword Recognition

Which of the following is a Java keyword?

A. println

B. void

C. main

D. System

Ans: B

Explanation: *'void' is a reserved keyword used to indicate that a method returns no value.*

Q5 | Identifiers and Keywords | Naming Rule

Which statement about Java identifier naming rules is TRUE?

A. An identifier can start with a digit
B. An identifier can be the same as a reserved keyword
C. An identifier can start with an underscore
D. An identifier can contain spaces

Ans: C

Explanation: *Java allows identifiers to begin with a letter, underscore (_), or dollar sign ($), but not a digit or space.*

Q6 | Identifiers and Keywords | Case Sensitivity

Java identifiers are case-sensitive. Which pair represents two DIFFERENT identifiers?

A. total and Total
B. num and num
C. sum and sum
D. data and data

Ans: A

Explanation: *Since Java is case-sensitive, 'total' and 'Total' are treated as two distinct identifiers.*

Q7 | Identifiers and Keywords | Keyword Recognition

Which of the following is NOT a Java keyword?

A. final
B. Boolean
C. this
D. static

Ans: B

Explanation: *'boolean' (lowercase) is a keyword, but 'Boolean' (capital B) is only a class name, not a reserved keyword.*

Q8 | Identifiers and Keywords | Naming Rule

Which of the following is a valid Java identifier?

A. 2ndValue
B. my value
C. class
D. $total

Ans: D

Explanation: *Java allows identifiers to begin with a dollar sign ($); '$total' follows all naming rules.*

Q9 | Identifiers and Keywords | Identifier vs Keyword

Which of these can legally be used as a variable name, even though it is poor practice?

A. class
B. String
C. void
D. int

Ans: B

Explanation: *'String' is a class name, not a reserved keyword, so it is technically a legal (though discouraged) identifier.*

Q10 | Identifiers and Keywords | Rule

Which of the following statements about Java keywords is TRUE?

A. Keywords cannot be used as identifiers
B. Keywords can start with a digit
C. Keywords can be freely used as variable names
D. Keywords are optional in Java programs

Ans: A

Explanation: *Keywords are reserved by the language and can never be used as identifiers such as variable or class names.*

Q11 | Identifiers and Keywords | Naming Rule

Which of the following identifiers is INVALID?

A. student1
B. student_1
C. _student1
D. student#1

Ans: D

Explanation: *The '#' symbol is not permitted in Java identifiers; only letters, digits, underscore, and dollar sign are allowed.*

Q12 | Identifiers and Keywords | Identification

Which word in the list below is a reserved Java keyword?

A. switchCase
B. Switch
C. switch
D. SWITCH

Ans: C

Explanation: *'switch' in lowercase is a reserved keyword used for the switch decision-making statement.*

Q13 | Identifiers and Keywords | Naming Rule

What is the maximum restriction Java places on the FIRST character of an identifier?

A. It cannot be a letter
B. It cannot be a dollar sign
C. It cannot be a digit
D. It cannot be an underscore

Ans: C

Explanation: *Java identifiers may start with a letter, underscore, or dollar sign, but never with a digit.*

Q14 | Identifiers and Keywords | Keyword Recognition

Which of the following is a valid Java keyword related to control flow?

A. BREAK
B. Break
C. breakLoop
D. break

Ans: D

Explanation: *'break' is a reserved keyword used to exit loops and switch statements.*

Q15 | Identifiers and Keywords | Naming Convention

Which identifier correctly represents a constant using standard Java rules?

A. MAX_VALUE

B. MAX VALUE

C. 1MAX

D. Max-Value

Ans: A

Explanation: *MAX_VALUE uses only letters and underscore and does not start with a digit, so it is a valid identifier.*

Q16 | Identifiers and Keywords | Rule-Based

Why is 'float' an invalid identifier in Java?

A. It contains an invalid character

B. It is a reserved keyword for a primitive data type

C. It exceeds the maximum identifier length

D. It starts with a digit

Ans: B

Explanation: *'float' is reserved as a keyword denoting a primitive data type, so it cannot be reused as an identifier.*

Q17 | Identifiers and Keywords | Error Identification

Identify the incorrect declaration based on identifier rules.

A. int second = 5;

B. int second2 = 5;

C. int _second = 5;

D. int 2nd = 5;

Ans: D

Explanation: *'2nd' is invalid because identifiers cannot begin with a digit.*

Q18 | Identifiers and Keywords | Conceptual

Which of the following BEST distinguishes a keyword from an identifier?

A. A keyword is longer than an identifier

B. A keyword can be changed by the programmer

C. A keyword has a predefined meaning in Java; an identifier is a name chosen by the programmer

D. An identifier must start with a capital letter

Ans: C

Explanation: *Keywords are fixed, reserved words with special meaning to the compiler, while identifiers are user-chosen names.*

Q19 | Identifiers and Keywords | Naming Rule

Which of the following identifiers is valid?

A. marks 2024

B. _marks2024

C. marks-2024

D. 2024marks

Ans: B

Explanation: *An identifier may start with an underscore and contain digits after the first character, making '_marks2024' valid.*

Q20 | Identifiers and Keywords | Assertion-Reason

Assertion (A): 'class' cannot be used as a variable name. Reason (R): 'class' is a reserved keyword in Java.

A. Both A and R are true, and R correctly explains A

B. A is false but R is true

C. Both A and R are false

D. A is true but R is false

Ans: A

Explanation: *'class' is indeed a reserved keyword, which is exactly why it cannot be used as a variable name.*

Section 2 — Data Types in Java

Q21 | Data Types in Java | Concept

What is the primary purpose of a data type in Java?

A. To control the flow of a program

B. To define the name of a variable

C. To specify the visibility of a class

D. To specify the kind of value a variable can hold and the operations allowed on it

Ans: D

Explanation: *A data type tells the compiler what kind of value a variable stores and what operations are valid on it.*

Q22 | Data Types in Java | Primitive Types

Which of the following is NOT a primitive data type in Java?

A. String

B. boolean

C. char

D. int

Ans: A

Explanation: *String is a reference (class) type in Java, not one of the eight primitive data types.*

Q23 | Data Types in Java | Integer Types

Which primitive data type occupies the LEAST memory in Java?

A. short

B. byte

C. long

D. int

Ans: B

Explanation: *byte is stored in 1 byte (8 bits), making it the smallest of Java's integer primitive types.*

Q24 | Data Types in Java | Integer Types

What is the size of the 'int' data type in Java?

A. 1 byte (8 bits)

B. 8 bytes (64 bits)

C. 4 bytes (32 bits)

D. 2 bytes (16 bits)

Ans: C

Explanation: *Java's int type is a fixed 32-bit signed integer regardless of the platform.*

Q25 | Data Types in Java | Floating-Point

Which data type is used to store a decimal number with double precision?

A. double
B. int
C. long
D. float

Ans: A

Explanation: *'double' provides double-precision (64-bit) floating-point storage, offering more precision than 'float'.*

Q26 | Data Types in Java | Character Type

How much memory does the 'char' data type occupy in Java?

A. 8 bytes
B. 4 bytes
C. 2 bytes
D. 1 byte

Ans: C

Explanation: *Java's char is 2 bytes because it uses Unicode encoding to represent a wide range of characters.*

Q27 | Data Types in Java | Boolean Type

What are the only two possible values of a boolean variable in Java?

A. yes and no
B. high and low
C. 0 and 1
D. true and false

Ans: D

Explanation: *A boolean variable in Java can only hold the literal values true or false, not numeric 0/1.*

Q28 | Data Types in Java | Default Values

What is the default value of an int instance variable in Java?

A. 1
B. 0
C. undefined
D. null

Ans: B

Explanation: *Uninitialized int instance variables are automatically set to 0 by the JVM.*

Q29 | Data Types in Java | Default Values

What is the default value of a boolean instance variable in Java?

A. true
B. 0
C. false
D. null

Ans: C

Explanation: *Boolean instance variables default to false when not explicitly initialized.*

Q30 | Data Types in Java | Range

Which data type would be MOST appropriate to store a person's age (e.g., 0-120)?

A. boolean
B. double
C. long
D. byte

Ans: D

Explanation: *Since byte can hold values from -128 to 127, it comfortably and efficiently stores typical age values.*

Q31 | Data Types in Java | Range

Which data type should be used to store very large whole numbers beyond the range of int?

A. byte
B. long
C. short
D. float

Ans: B

Explanation: *'long' is a 64-bit integer type designed to hold much larger values than int can accommodate.*

Q32 | Data Types in Java | Declaration

Which of the following is a correct declaration of a float variable in Java?

A. float price = 99.5f;
B. float price = 99.5;
C. float price = 99.5d;
D. float price = "99.5";

Ans: A

Explanation: *A float literal must be suffixed with 'f' or 'F'; without it, the value is treated as a double by default.*

Q33 | Data Types in Java | Declaration

Which declaration correctly stores a single character?

A. char grade = A;
B. char grade = "A";
C. char grade = 65.0;
D. char grade = 'A';

Ans: D

Explanation: *Character literals in Java must be enclosed in single quotes, not double quotes.*

Q34 | Data Types in Java | Type Identification

Which data type would be most suitable to represent whether a student has passed or failed?

A. byte
B. boolean
C. String
D. char

Ans: B

Explanation: *A pass/fail status has only two possible states, which boolean directly represents as true/false.*

Q35 | Data Types in Java | Comparison

Which of the following correctly orders integer types from SMALLEST to LARGEST range?

A. short, byte, long, int

B. int, byte, short, long

C. byte, short, int, long

D. long, int, short, byte

Ans: C

Explanation: *Java's integer types increase in size and range in the order: byte (1B) < short (2B) < int (4B) < long (8B).*

Q36 | Data Types in Java | Code-Based

What will be the default value of a double instance variable that is declared but not initialized?

A. 0.0

B. NaN

C. 0

D. null

Ans: A

Explanation: *Uninitialized double instance variables default to 0.0, not integer 0.*

Q37 | Data Types in Java | Conceptual

Which of the following is classified as a reference data type in Java (at introductory level)?

A. Array

B. boolean

C. char

D. int

Ans: A

Explanation: *Arrays are reference types in Java because array variables store a reference to the array object, unlike primitives.*

Q38 | Data Types in Java | Range/Precision

Which data type provides greater precision for decimal calculations, float or double?

A. Both provide equal precision

B. Neither supports decimals

C. double

D. float

Ans: C

Explanation: *double uses 64 bits and offers roughly twice the precision of the 32-bit float type.*

Q39 | Data Types in Java | Type Selection

Which data type is most appropriate for storing a bank account balance requiring high decimal precision?

A. char

B. double

C. byte

D. boolean

Ans: B

Explanation: *double offers sufficient precision and range for monetary values with decimals, unlike small integer types.*

Q40 | Data Types in Java | Error Identification

Identify the error in the following declaration:

```
int marks = 99.5;
```

A. Semicolon is missing

B. 'marks' is an invalid identifier

C. int cannot store numeric values

D. A decimal value cannot be assigned directly to an int variable

Ans: D

Explanation: *99.5 is a floating-point literal; assigning it to an int without explicit casting causes a compile-time error.*

Section 3 — Java Coding Conventions

Q41 | Java Coding Conventions | Class Naming

According to standard Java coding conventions, how should class names be written?

A. Starting with an uppercase letter, using CamelCase (e.g., StudentRecord)

B. Starting with an underscore (e.g., _StudentRecord)

C. Entirely in lowercase (e.g., studentrecord)

D. Entirely in uppercase (e.g., STUDENTRECORD)

Ans: A

Explanation: *Java convention requires class names to begin with an uppercase letter and follow CamelCase style.*

Q42 | Java Coding Conventions | Method Naming

Which of the following method names follows standard Java naming conventions?

A. calculate_total()

B. CALCULATETOTAL()

C. calculateTotal()

D. CalculateTotal()

Ans: C

Explanation: *Method names should start with a lowercase letter and use camelCase, as in calculateTotal().*

Q43 | Java Coding Conventions | Variable Naming

Which variable name follows Java coding conventions?

A. StudentAge

B. studentAge

C. student_age_value_123abc

D. STUDENTAGE

Ans: B

Explanation: *Variable names conventionally start with a lowercase letter and use camelCase for readability.*

Q44 | Java Coding Conventions | Constant Naming

How should constants typically be named in Java according to convention?

A. With a leading dollar sign, e.g., $MAXSPEED

B. In PascalCase, e.g., MaxSpeed

C. In camelCase, e.g., maxSpeed

D. In uppercase letters with underscores, e.g., MAX_SPEED

Ans: D

Explanation: *Constants (declared with 'final') are conventionally written in uppercase with underscores separating words.*

Q45 | Java Coding Conventions | Formatting

Why is consistent indentation considered an important Java coding convention?

A. It is required by the Java compiler to run the code

B. It reduces the size of the compiled program

C. It increases the execution speed of the program

D. It improves code readability and helps show the logical structure of the program

Ans: D

Explanation: *Indentation is purely a readability practice; it has no effect on compilation or performance.*

Q46 | Java Coding Conventions | General Convention

Which of these best reflects good Java coding convention for package names?

A. All uppercase letters, e.g., COM.SCHOOL.APP

B. All lowercase letters, e.g., com.school.app

C. Mixed with underscores, e.g., com_school_app

D. CamelCase, e.g., Com.School.App

Ans: B

Explanation: *Java package names are conventionally written entirely in lowercase.*

Q47 | Java Coding Conventions | Naming Convention

Which of the following class names correctly follows Java naming conventions?

A. BankAccount

B. BANKACCOUNT

C. bank_account

D. bankAccount

Ans: A

Explanation: *Class names should begin with an uppercase letter and follow CamelCase, as in BankAccount.*

Q48 | Java Coding Conventions | Readability

Which coding practice is recommended by Java conventions to improve readability of long methods?

A. Removing all comments to reduce file size

B. Using single-letter variable names throughout

C. Breaking the method into smaller, well-named methods

D. Writing the entire logic in a single line

Ans: C

Explanation: *Breaking code into smaller, clearly named methods is a widely recommended convention for readability.*

Q49 | Java Coding Conventions | Comments

According to Java coding conventions, comments should primarily be used to:

A. Replace the need for meaningful variable names

B. Explain the purpose or logic of code that is not immediately obvious

C. Increase the file size for documentation purposes

D. Disable code permanently instead of deleting it

Ans: B

Explanation: *Comments are meant to clarify intent and logic, not substitute for clear naming or serve as permanent code disablers.*

Q50 | Java Coding Conventions | Naming Convention

Which boolean variable name best follows Java naming conventions?

A. isAvailable

B. available_Flag

C. Isavailable

D. IS_AVAILABLE_FLAG

Ans: A

Explanation: *Boolean variables are conventionally prefixed with 'is' or 'has' and follow camelCase, e.g., isAvailable.*

Q51 | Java Coding Conventions | Application

A programmer names a class 'student_record'. Which convention does this VIOLATE?

A. Class names cannot contain the letter 'r'

B. Class names must be written in a single case only

C. Class names should use CamelCase starting with an uppercase letter

D. Class names must be a single word

Ans: C

Explanation: *Using lowercase with underscores instead of CamelCase with a leading capital breaks Java's class naming convention.*

Q52 | Java Coding Conventions | Consistency

Why do Java coding conventions recommend a CONSISTENT coding style across a project?

A. It makes the compiled .class file smaller

B. It automatically improves the program's runtime speed

C. It is mandatory for the Java compiler to accept the code

D. It makes code easier to read, maintain, and collaborate on across a team

Ans: D

Explanation: *Consistent style is a readability and maintainability practice; it has no effect on compilation or runtime behavior.*

Section 4 — Expressions in Java

Q53 | Expressions in Java | Concept

What is a Java expression?

A. A combination of variables, operators, and values that evaluates to a single value

B. A statement that always ends with a semicolon and does nothing else

C. A keyword used to define a class

D. A block of code enclosed in curly braces

Ans: A

Explanation: *An expression combines operands and operators and produces a resulting value when evaluated.*

Q54 | Expressions in Java | Components

In the expression 'a + b', what are 'a' and 'b' called?

A. Identifiers only

B. Operands

C. Literals only

D. Operators

Ans: B

Explanation: *The values or variables that an operator acts upon are called operands.*

Q55 | Expressions in Java | Arithmetic

What is the result of the expression 10 % 3 in Java?

A. 0

B. 3.33

C. 1

D. 3

Ans: C

Explanation: *The modulus operator (%) returns the remainder of division; 10 divided by 3 leaves a remainder of 1.*

Q56 | Expressions in Java | Arithmetic

What is the output of the following expression?

```
int result = 7 / 2;
```

A. 4

B. 1

C. 3.5

D. 3

Ans: D

Explanation: *Since both operands are int, integer division is performed and the fractional part is discarded, giving 3.*

Q57 | Expressions in Java | Relational

What does the expression (5 > 3) evaluate to?

A. 1

B. true

C. false

D. 5

Ans: B

Explanation: *Relational operators like > always evaluate to a boolean result; since 5 is greater than 3, the result is true.*

Q58 | Expressions in Java | Logical

What is the result of the expression (true && false)?

A. 1

B. true

C. 0

D. false

Ans: D

Explanation: *The logical AND (&&) operator returns true only if both operands are true; here one operand is false.*

Q59 | Expressions in Java | Logical

What is the result of the expression (false || true)?

A. true

B. 0

C. null

D. false

Ans: A

Explanation: *The logical OR (||) operator returns true if at least one operand is true.*

Q60 | Expressions in Java | Assignment

What is the value of x after executing the following?

```
int x = 5;
x += 3;
```

A. 3

B. 15

C. 8

D. 5

Ans: C

Explanation: *The compound assignment operator += adds the right operand to x, so x becomes 5 + 3 = 8.*

Q61 | Expressions in Java | Unary

What is the value of x after the following code executes?

```
int x = 5;
x++;
```

A. 6

B. 5

C. 4

D. 7

Ans: A

Explanation: *The post-increment operator (++) increases the value of x by 1, changing it from 5 to 6.*

Q62 | Expressions in Java | Unary

What does the unary minus operator do in the expression '-x'?

A. It converts x into a negative index

B. It negates the value of x

C. It divides x by -1 permanently

D. It decreases x by 1

Ans: B

Explanation: *The unary minus operator simply reverses the sign of the operand's value.*

Q63 | Expressions in Java | Precedence

In the expression '10 + 5 * 2', which operation is performed FIRST?

A. The whole expression left to right

B. 2 is evaluated alone

C. 10 + 5

D. 5 * 2

Ans: D

Explanation: *Multiplication has higher precedence than addition, so 5 * 2 is evaluated before the addition.*

Q64 | Expressions in Java | Precedence

What is the result of the expression: 20 - 4 / 2?

A. 10

B. 16

C. 18

D. 8

Ans: C

Explanation: *Division has higher precedence than subtraction: 4 / 2 = 2, then 20 - 2 = 18.*

Q65 | Expressions in Java | Associativity

In the expression 'a = b = c = 5;', in which direction does the assignment operator associate?

A. Right to left

B. It cannot be chained in Java

C. Left to right

D. Top to bottom only

Ans: A

Explanation: *The assignment operator (=) is right-associative, so c is assigned first, then b, then a, all receiving 5.*

Q66 | Expressions in Java | Evaluation

What is the output of the following code?

```
int a = 6, b = 4;
System.out.println(a > b ? "Yes" : "No");
```

A. No

B. 6

C. true

D. Yes

Ans: D

Explanation: *The ternary operator evaluates the condition a > b (true), so it returns the value before the colon, "Yes".*

Q67 | Expressions in Java | Type-Related

What is the result of the expression: 5 + 2.0 in Java?

A. 2.0

B. 5.2

C. 7.0

D. 7

Ans: C

Explanation: *When an int is combined with a double using +, the int is implicitly promoted to double, producing 7.0.*

Q68 | Expressions in Java | Expression Identification

Which of the following is a valid Java expression?

A. int a;

B. a + b * c

C. class Test {}

D. if (a > b)

Ans: B

Explanation: *'a + b * c' combines operands with operators and produces a value, making it a true expression, unlike a declaration or statement.*

Q69 | Expressions in Java | Output-Based

What is printed by the following code?

```
int a = 15;
System.out.println(a % 4);
```

A. 3

B. 15

C. 0

D. 4

Ans: A

Explanation: *15 divided by 4 gives a quotient of 3 with a remainder of 3, so a % 4 evaluates to 3.*

Q70 | Expressions in Java | Relational/Logical

What is the output of the following?

```
int x = 5;
System.out.println(x != 5);
```

A. 0

B. true

C. false

D. 5

Ans: C

Explanation: *The != operator checks inequality; since x equals 5, the comparison x != 5 evaluates to false.*

Q71 | Expressions in Java | Correct/Incorrect

Which of the following expressions is INCORRECT in Java?

A. int x = 5 % 6;

B. int x = 5 = 6;

C. int x = (5 + 6) * 2;

D. int x = 5 + 6;

Ans: B

Explanation: *'5 = 6' attempts to assign a value to a literal, which is illegal because a literal is not a variable.*

Q72 | Expressions in Java | Operator-Based

Which operator has the HIGHEST precedence among the following?

A. + (addition)

B. || (logical OR)

C. = (assignment)

D. * (multiplication)

Ans: D

Explanation: *Among these, multiplication has the highest precedence, followed by addition, then logical OR, then assignment.*

Q73 | Expressions in Java | Code-Snippet

What is the value of 'result' after execution?

```
int result = (2 + 3) * (4 - 1);
```

A. 15

B. 14

C. 9

D. 10

Ans: A

Explanation: *Parentheses are evaluated first: (2+3)=5 and (4-1)=3, then 5 * 3 = 15.*

Q74 | Expressions in Java | Evaluation

What is the output of the following code?

```
boolean flag = (10 == 10) && (5 < 3);
```

A. true

B. 10

C. 5

D. false

Ans: D

Explanation: *10==10 is true, but 5<3 is false; since && requires both to be true, the overall result is false.*

Section 5 — Control Structures & Decision-Making Statements

Q75 | Control Structures & Decision-Making Statements | Concept

What is the default flow of execution in a Java program without any control structures?

A. Determined by variable names

B. Random order chosen by the JVM

C. Sequential, from top to bottom

D. Bottom to top

Ans: C

Explanation: *Without decision-making or looping constructs, Java statements execute sequentially in the order written.*

Q76 | Control Structures & Decision-Making Statements | Concept

What is the main purpose of a decision-making statement in Java?

A. To define a new class

B. To alter the flow of execution based on a condition

C. To repeat a block of code indefinitely

D. To declare new variables

Ans: B

Explanation: *Decision-making statements like if and switch allow the program to choose between different execution paths.*

Q77 | Control Structures & Decision-Making Statements | if Statement

Which of the following is the correct syntax for an if statement in Java?

A. if [condition] { }

B. if condition { }

C. if (condition) { }

D. if condition then { }

Ans: C

Explanation: *Java requires the condition of an if statement to be enclosed in parentheses.*

Q78 | Control Structures & Decision-Making Statements | if-else

What happens if the condition in an if-else statement is false?

A. The program terminates

B. The else block is executed

C. Both blocks are executed

D. The if block is executed anyway

Ans: B

Explanation: *When the if condition evaluates to false, control passes to the else block, if present.*

Q79 | Control Structures & Decision-Making Statements | Output-Based

What is the output of the following code?

```
int a = 10;
if (a > 5)
System.out.println("High");
else
System.out.println("Low");
```

A. High

B. Low

C. 10

D. No output

Ans: A

Explanation: *Since a (10) is greater than 5, the condition is true and "High" is printed.*

Q80 | Control Structures & Decision-Making Statements | else-if Ladder

In an else-if ladder, how many conditions can be evaluated in total?

A. Exactly two

B. Unlimited, but only the last condition is checked

C. Only one

D. As many as the programmer defines using additional else if blocks

Ans: D

Explanation: *An else-if ladder can contain any number of else-if blocks, each checked in sequence until one is true.*

Q81 | Control Structures & Decision-Making Statements | Nested if

What is a nested if statement?

A. An if statement that has no condition

B. An if statement placed inside another if or else block

C. An if statement used only with arrays

D. A loop that repeats an if block

Ans: B

Explanation: *A nested if is simply an if statement written inside the body of another if or else statement.*

Q82 | Control Structures & Decision-Making Statements | Output-Based

What is the output of the following nested if code?

```
int a = 20;
if (a > 10) {
if (a > 15)
System.out.println("A");
else
System.out.println("B");
} else {
System.out.println("C");
}
```

A. C

B. No output

C. A

D. B

Ans: C

Explanation: *a (20) is greater than 10, entering the outer block; then a is also greater than 15, so "A" is printed.*

Q83 | Control Structures & Decision-Making Statements | switch

Which keyword is used to test multiple possible values of a variable in Java?

A. while

B. for

C. if

D. switch

Ans: D

Explanation: *The switch statement is designed to test a variable against multiple possible constant values (cases).*

Q84 | Control Structures & Decision-Making Statements | switch/case

What is the purpose of the 'break' statement inside a switch case?

A. To exit the switch block and prevent fall-through to subsequent cases

B. To skip the next iteration of a loop

C. To terminate the entire program

D. To restart the switch statement

Ans: A

Explanation: *Without break, execution 'falls through' into the next case; break stops this by exiting the switch block.*

Q85 | Control Structures & Decision-Making Statements | switch/default

What is the role of the 'default' label in a switch statement?

A. It always executes first regardless of matches

B. It stops the program if reached

C. It is mandatory in every switch statement

D. It executes when none of the case values match the switch expression

Ans: D

Explanation: *The default case runs only when no other case value matches; it is optional, not mandatory.*

Q86 | Control Structures & Decision-Making Statements | Output-Based

What is the output of the following code?

```
int day = 3;
switch (day) {
case 1: System.out.println("Mon"); break;
case 2: System.out.println("Tue"); break;
case 3: System.out.println("Wed"); break;
default: System.out.println("Invalid");
}
```

A. Tue

B. Invalid

C. Wed

D. Mon

Ans: C

Explanation: *Since day equals 3, case 3 matches and prints "Wed", then break exits the switch.*

Q87 | Control Structures & Decision-Making Statements | Fall-Through

What is the output of the following code (note the missing break)?

```
int x = 1;
switch (x) {
case 1: System.out.println("One");
case 2: System.out.println("Two"); break;
default: System.out.println("Other");
}
```

A. One Two

B. Two

C. Other

D. One

Ans: A

Explanation: *Without a break after case 1, execution falls through into case 2, printing both "One" and "Two".*

Q88 | Control Structures & Decision-Making Statements | Comparison

What is a key difference between if-else and switch statements?

A. switch can only be used with strings

B. switch compares a single variable against multiple discrete values, while if-else can test complex conditions

C. if-else cannot use relational operators

D. switch executes all cases every time

Ans: B

Explanation: *switch is best suited for matching one variable against fixed values; if-else handles more general boolean conditions.*

Q89 | Control Structures & Decision-Making Statements | Nested Decision

What is the output of the following code?

```
int marks = 45;
if (marks >= 90)
System.out.println("A");
else if (marks >= 60)
System.out.println("B");
else if (marks >= 40)
System.out.println("C");
else
System.out.println("D");
```

A. C

B. A

C. D

D. B

Ans: A

Explanation: *45 fails the first two conditions but satisfies marks >= 40, so "C" is printed.*

Q90 | Control Structures & Decision-Making Statements | Working

In an if-else-if ladder, how many blocks are executed if a matching condition is found in the middle?

A. The entire ladder repeats

B. All blocks after it are also executed

C. Only that one block, and the rest are skipped

D. All blocks before and after it execute

Ans: C

Explanation: *Once a condition in the ladder is found true, its block executes and the remaining else-if/else branches are skipped.*

Q91 | Control Structures & Decision-Making Statements | Condition Evaluation

What is the output of the following code?

```
int a = 7;
if (a % 2 == 0)
System.out.println("Even");
else
System.out.println("Odd");
```

A. 0

B. Odd

C. Even

D. 7

Ans: B

Explanation: *7 % 2 evaluates to 1, which is not 0, so the condition is false and "Odd" is printed.*

Q92 | Control Structures & Decision-Making Statements | Identifying Structure

Which control structure would be MOST appropriate to check a single variable against many fixed integer values?

A. while loop
B. for loop
C. do-while loop
D. switch statement

Ans: D

Explanation: *switch is specifically designed for matching one variable against several discrete constant values efficiently.*

Q93 | Control Structures & Decision-Making Statements | case Labels

Can two case labels in the same switch statement have the same value in Java?

A. Yes, always
B. Yes, but only in nested switches
C. No, it causes a compile-time error
D. Yes, but only for integer values

Ans: C

Explanation: *Duplicate case values in the same switch block are not allowed and result in a compilation error.*

Q94 | Control Structures & Decision-Making Statements | switch Data Types

Which of the following data types CANNOT be directly used as a switch expression in basic Java (from this syllabus)?

A. int
B. char
C. byte
D. boolean

Ans: D

Explanation: *switch works with types like byte, short, char, int, and String, but not boolean, since boolean has only two values better suited to if-else.*

Q95 | Control Structures & Decision-Making Statements | Output-Based

What is the output of the following code?

```
char grade = 'B';
switch (grade) {
case 'A': System.out.println("Excellent"); break;
case 'B': System.out.println("Good"); break;
case 'C': System.out.println("Average"); break;
}
```

A. Good
B. Average
C. No output
D. Excellent

Ans: A

Explanation: *The switch expression grade holds 'B', which matches case 'B', printing "Good".*

Q96 | Control Structures & Decision-Making Statements | Sequential Execution

Which of the following best describes sequential execution?

A. Statements are executed based on user input order

B. Statements are executed one after another in the order they appear

C. Statements are executed in reverse order

D. Statements are executed randomly

Ans: B

Explanation: *Sequential execution simply means each statement runs in the exact order it is written, unless altered by control statements.*

Q97 | Control Structures & Decision-Making Statements | Nested if Output

What is the output of the following code?

```
int x = 5, y = 10;
if (x > 0) {
if (y > 5)
System.out.println("Both");
}
else
System.out.println("None");
```

A. x > 0

B. None

C. Both

D. No output

Ans: C

Explanation: *x > 0 is true, entering the outer if; y > 5 is also true, so the inner if prints "Both".*

Q98 | Control Structures & Decision-Making Statements | break Behavior

What happens if 'break' is omitted from EVERY case in a switch statement, and the matched case is not the last one?

A. The program skips the entire switch block

B. Only the matched case executes and the switch exits automatically

C. The switch statement throws a compile-time error

D. Execution continues into all subsequent cases until a break or the end of the switch

Ans: D

Explanation: *Without break statements, Java falls through and executes all subsequent case blocks in sequence.*

Section 6 — Arrays and Their Methods

Q99 | Arrays and Their Methods | Concept

What is an array in Java?

A. A type of loop used for repeated execution

B. A collection of elements of the same data type stored in contiguous memory under a single name

C. A collection of methods grouped together

D. A single variable that can store only one value

Ans: B

Explanation: *An array groups multiple values of the same type together, accessible through a common variable name and index.*

Q100 | Arrays and Their Methods | Purpose

What is the main advantage of using an array instead of separate variables?

A. It allows storing and managing multiple related values efficiently using a single variable name

B. It converts data types automatically

C. It removes the need for loops entirely

D. It automatically sorts the values

Ans: A

Explanation: *Arrays let a programmer manage many values of the same type together instead of declaring separate variables.*

Q101 | Arrays and Their Methods | Declaration

Which of the following correctly declares an array of integers in Java?

A. int[] numbers;

B. int numbers[5];

C. int numbers = [];

D. array int numbers;

Ans: A

Explanation: *'int[] numbers;' is valid Java array declaration syntax; the size is specified during array creation, not declaration.*

Q102 | Arrays and Their Methods | Creation

Which statement correctly creates an array of 5 integers?

A. int[] numbers = new int(5);

B. int[] numbers = int[5];

C. int[] numbers = new int[5];

D. int[5] numbers = new int();

Ans: C

Explanation: *The 'new' keyword followed by the type and size in square brackets correctly allocates array memory.*

Q103 | Arrays and Their Methods | Initialization

Which of the following correctly initializes an array with values at the time of declaration?

A. int[] arr = new int[]{10, 20, 30]

B. int[] arr = [10, 20, 30];

C. int arr[] = (10, 20, 30);

D. int[] arr = {10, 20, 30};

Ans: D

Explanation: *Curly braces {} are used for array initializer lists in Java when declaring and assigning values together.*

Q104 | Arrays and Their Methods | Indexing

In Java, what is the index of the FIRST element of an array?

A. 1

B. 0

C. It depends on the array size

D. -1

Ans: B

Explanation: *Java arrays are zero-indexed, meaning the first element is always accessed using index 0.*

Q105 | Arrays and Their Methods | Length

Which of the following correctly retrieves the number of elements in an array named 'arr'?

A. arr.length

B. arr.size()

C. arr.size

D. arr.length()

Ans: A

Explanation: *For arrays, 'length' is a property (not a method), accessed without parentheses, unlike String's length() method.*

Q106 | Arrays and Their Methods | Accessing Elements

What is the output of the following code?

```
int[] arr = {5, 10, 15, 20};
System.out.println(arr[2]);
```

A. 10

B. 5

C. 15

D. 20

Ans: C

Explanation: *Index 2 refers to the third element (since indexing starts at 0), which is 15.*

Q107 | Arrays and Their Methods | Array Length

What is the output of the following code?

```
int[] arr = {1, 2, 3, 4, 5};
System.out.println(arr.length);
```

A. 6

B. 4

C. 0

D. 5

Ans: D

Explanation: *The array contains 5 elements, so arr.length returns 5.*

Q108 | Arrays and Their Methods | Traversal

Which loop construct is most commonly used to traverse all elements of an array?

A. single System.out.println statement

B. for loop

C. switch statement

D. if statement

Ans: B

Explanation: *A for loop, often using arr.length as the bound, is the standard way to iterate through every array element.*

Q109 | Arrays and Their Methods | Error Identification

What is wrong with the following code?

```
int[] arr = new int[5];
System.out.println(arr[5]);
```

A. It causes an ArrayIndexOutOfBoundsException since valid indices are 0 to 4

B. It automatically resizes the array

C. It prints 0 successfully

D. It causes a compile-time error

Ans: A

Explanation: *An array of size 5 has valid indices 0 through 4; accessing index 5 is out of bounds and throws a runtime exception.*

Q110 | Arrays and Their Methods | Default Values

What is the default value of each element in a newly created int array?

A. Undefined/garbage value

B. 1

C. null

D. 0

Ans: D

Explanation: *Java automatically initializes numeric array elements to 0 (and boolean to false, object references to null).*

Q111 | Arrays and Their Methods | Multidimensional

Which of the following declares a basic two-dimensional array in Java?

A. int matrix[3,3] = new int();

B. int[][] matrix = new int[3][3];

C. int[] matrix = new int[3,3];

D. int[3][3] matrix = new int();

Ans: B

Explanation: *A 2D array is declared using two sets of square brackets, as in int[][] matrix = new int[3][3];.*

Q112 | Arrays and Their Methods | Output-Based

What is the output of the following code?

```
int[] arr = {2, 4, 6, 8};
int sum = 0;
for (int i = 0; i < arr.length; i++) {
sum += arr[i];
}
System.out.println(sum);
```

A. 24

B. 8

C. 20

D. 10

Ans: C

Explanation: *The loop adds all elements: 2 + 4 + 6 + 8 = 20.*

Q113 | Arrays and Their Methods | Array Elements

What is the value of arr[3] in the following declaration?

```
int[] arr = {11, 22, 33, 44, 55};
```

A. 55

B. 44

C. 11

D. 33

Ans: B

Explanation: *Index 3 refers to the fourth element in the array (0-indexed), which is 44.*

Q114 | Arrays and Their Methods | Basic Operations

Which of the following correctly updates the value at index 2 of an existing array 'arr' to 100?

A. arr[2] = 100;

B. arr(2) = 100;

C. arr.set(2, 100);

D. arr.2 = 100;

Ans: A

Explanation: *Array elements are accessed and modified using square-bracket index notation, as in arr[2] = 100;.*

Q115 | Arrays and Their Methods | Traversal Output

What is the output of the following code?

```
int[] arr = {1, 2, 3};
for (int i = 0; i < arr.length; i++) {
System.out.print(arr[i] + " ");
}
```

A. 3 2 1

B. 1 2 3 4

C. 0 1 2

D. 1 2 3

Ans: D

Explanation: *The loop prints each element from index 0 to length-1 in order, producing '1 2 3 '.*

Q116 | Arrays and Their Methods | Array Length Method

How is the length of an array 'names' correctly accessed in Java?

A. names.length()

B. length(names)

C. names.length

D. names.count

Ans: C

Explanation: *Unlike String's length() method, an array's length is accessed as a public field without parentheses.*

Q117 | Arrays and Their Methods | Concept

Which of the following statements about Java arrays is TRUE?

A. Array indices in Java start from 1

B. Once created, the size of an array cannot be changed

C. Arrays automatically grow in size when new elements are added

D. Arrays can store elements of different data types by default

Ans: B

Explanation: *Java arrays have a fixed size once created; a new array must be created to hold more elements.*

Q118 | Arrays and Their Methods | Output-Based

What is the output of the following code?

```
int[] arr = new int[4];
System.out.println(arr[0]);
```

A. Compilation error

B. Garbage value

C. null

D. 0

Ans: D

Explanation: *An int array's elements are automatically initialized to 0, so arr[0] prints 0 even without explicit assignment.*

Q119 | Arrays and Their Methods | Array Traversal

What is the output of the following code that finds the largest element?

```
int[] arr = {4, 9, 2, 7};
int max = arr[0];
for (int i = 1; i < arr.length; i++) {
if (arr[i] > max)
max = arr[i];
}
System.out.println(max);
```

A. 4

B. 2

C. 9

D. 7

Ans: C

Explanation: *The loop compares each element with the current max; 9 is the largest value in the array, so it is printed.*

Section 7 — Code and Output-Based Questions

Q120 | Code and Output-Based Questions | Mixed: Identifiers+Types

What is the output of the following code?

```
int age = 25;
String name = "OAVS";
System.out.println(name + age);
```

A. OAVS25

B. 25OAVS

C. OAVS + 25

D. Compilation error

Ans: A

Explanation: *When a String is concatenated with an int using +, Java converts the int to its string form and joins them: "OAVS" + 25 = "OAVS25".*

Q121 | Code and Output-Based Questions | Mixed: Expressions+Control

What is the output of the following code?

```
int a = 4, b = 5;
if ((a + b) % 2 == 0)
System.out.println("Even Sum");
else
System.out.println("Odd Sum");
```

A. 9

B. Even Sum

C. 0

D. Odd Sum

Ans: D

Explanation: *a + b = 9, and 9 % 2 = 1 (not 0), so the condition is false and "Odd Sum" is printed.*

Q122 | Code and Output-Based Questions | Mixed: Arrays+Loops

What is the output of the following code?

```
int[] arr = {3, 6, 9};
int product = 1;
for (int i = 0; i < arr.length; i++)
product *= arr[i];
System.out.println(product);
```

A. 9

B. 18

C. 162

D. 27

Ans: C

Explanation: *product accumulates as 1*3=3, 3*6=18, 18*9=162, so the final printed value is 162.*

Q123 | Code and Output-Based Questions | Mixed: switch+Arrays

What is the output of the following code?

```
int[] scores = {70, 85, 90};
int s = scores[1];
switch (s / 10) {
case 8: System.out.println("Grade B"); break;
case 9: System.out.println("Grade A"); break;
default: System.out.println("Grade C");
}
```

A. Grade B

B. Grade A

C. 85

D. Grade C

Ans: A

Explanation: *scores[1] is 85; 85 / 10 (integer division) equals 8, matching case 8 and printing "Grade B".*

Q124 | Code and Output-Based Questions | Mixed: Data Types+Expressions

What is the output of the following code?

```
double avg = (int) (15 / 4);
System.out.println(avg);
```

A. 3

B. 3.0

C. 4.0

D. 3.75

Ans: B

Explanation: *15 / 4 performs integer division first, giving 3; casting to int keeps it 3, and assigning to a double prints 3.0.*

Q125 | Code and Output-Based Questions | Mixed: Nested if+Arrays

What is the output of the following code?

```
int[] nums = {12, 45, 7};
if (nums[0] < nums[1]) {
if (nums[2] < nums[0])
System.out.println("X");
else
System.out.println("Y");
} else {
System.out.println("Z");
}
```

A. No output

B. X

C. Z

D. Y

Ans: B

Explanation: *nums[0]=12 is less than nums[1]=45, so the outer if runs; nums[2]=7 is less than nums[0]=12, so the inner if is true and "X" is printed.*

Q126 | Code and Output-Based Questions | Mixed: Coding Convention+Types

Which of the following variable declarations follows BOTH correct Java data type usage and naming conventions?

A. int 95marks = totalMarks;

B. Int Total_Marks = 95;

C. int totalMarks = 95;

D. int TOTALMARKS = 95;

Ans: C

Explanation: *'int totalMarks = 95;' uses a valid lowercase primitive type keyword and a camelCase identifier as per convention.*

Q127 | Code and Output-Based Questions | Mixed: Loops+Expressions

What is the output of the following code?

```
int count = 0;
for (int i = 1; i <= 5; i++) {
if (i % 2 == 0)
count++;
}
System.out.println(count);
```

A. 2

B. 4

C. 3

D. 5

Ans: A

Explanation: *Among 1 to 5, the even numbers are 2 and 4, so count is incremented twice, resulting in 2.*

Q128 | Code and Output-Based Questions | Mixed: Arrays+Expressions

What is the output of the following code?

```
int[] arr = {10, 20, 30};
System.out.println(arr[0] + arr[arr.length - 1]);
```

A. 60

B. 50

C. 30

D. 40

Ans: D

Explanation: *arr.length - 1 is 2, so arr[2] is 30; arr[0] is 10; 10 + 30 = 40.*

Q129 | Code and Output-Based Questions | Mixed: switch fall-through

What is the output of the following code?

```
int n = 2;
switch (n) {
case 1:
case 2:
case 3:
System.out.println("Low");
break;
default:
System.out.println("High");
}
```

A. High

B. Low

C. 1

D. 3

Ans: B

Explanation: *Cases 1, 2, and 3 share the same block (grouped cases); since n is 2, it matches and prints "Low".*

Q130 | Code and Output-Based Questions | Mixed: Data Types+Arrays

What is the output of the following code?

```
char[] letters = {'J', 'A', 'V', 'A'};
System.out.println(letters[1]);
```

A. J

B. V

C. letters[1]

D. A

Ans: D

Explanation: *Index 1 in the char array refers to the second element, which is 'A'.*

Q131 | Code and Output-Based Questions | Mixed: Expressions+if-else

What is the output of the following code?

```
int x = 3, y = 3;
System.out.println(x == y ? "Equal" : "Not Equal");
```

A. Equal

B. true

C. 3

D. Not Equal

Ans: A

Explanation: *x == y evaluates to true since both are 3, so the ternary operator returns "Equal".*

Q132 | Code and Output-Based Questions | Mixed: Loops+Arrays

What is the output of the following code?

```
int[] arr = {5, 10, 15, 20, 25};
int count = 0;
for (int i = 0; i < arr.length; i++) {
if (arr[i] > 10)
count++;
}
System.out.println(count);
```

A. 4

B. 2

C. 3

D. 5

Ans: C

Explanation: *Elements greater than 10 are 15, 20, and 25 — three elements in total, so count becomes 3.*

Q133 | Code and Output-Based Questions | Mixed: Identifiers+Expressions

Which of the following code snippets contains an ERROR?

A. int total1 = a + b;

B. int Total1 = a + b;

C. int total_1 = a + b;

D. int 1total = a + b;

Ans: D

Explanation: *'1total' is an invalid identifier because it begins with a digit, causing a compile-time error.*

Q134 | Code and Output-Based Questions | Mixed: Control+Data Types

What is the output of the following code?

```
boolean isPass = true;
int marks = 35;
if (isPass && marks >= 40)
System.out.println("Pass");
else
System.out.println("Fail");
```

A. true

B. 35

C. Fail

D. Pass

Ans: C

Explanation: *Although isPass is true, marks >= 40 is false (35 < 40); since && needs both true, the result is "Fail".*

Q135 | Code and Output-Based Questions | Mixed: Arrays+switch

What is the output of the following code?

```
int[] vals = {1, 2, 3};
for (int v : vals) {
switch (v) {
case 2: System.out.print("Two"); break;
default: System.out.print("-");
}
}
```

A. -Two-

B. Two--

C. --Two

D. TwoTwoTwo

Ans: A

Explanation: *The enhanced for loop visits 1, 2, 3 in order; only 2 matches case 2 ('Two'), while 1 and 3 fall to default ('-'), giving "-Two-".*

Section 8 — Difficult & Tricky Questions

Q136 | Difficult & Tricky Questions | Tricky: Type Promotion

What is the output of the following code?

```
byte b = 10;
b = b + 5;
System.out.println(b);
```

A. 10

B. Compilation error

C. 5

D. 15

Ans: B

Explanation: *b + 5 is promoted to int automatically; assigning an int back to a byte without an explicit cast causes a compile-time error.*

Q137 | Difficult & Tricky Questions | Tricky: Operator Precedence

What is the output of the following code?

```
int a = 5;
int b = a++ + ++a;
System.out.println(b);
```

A. 13

B. 12

C. 11

D. 10

Ans: B

Explanation: *a++ uses 5 then makes a=6; ++a makes a=7 then uses 7; so b = 5 + 7 = 12.*

Q138 | Difficult & Tricky Questions | Tricky: switch without break

What is the output of the following code?

```
int x = 2;
switch (x) {
case 1: System.out.print("A");
case 2: System.out.print("B");
case 3: System.out.print("C"); break;
case 4: System.out.print("D");
}
```

A. ABC

B. BCD

C. B

D. BC

Ans: D

Explanation: *Execution starts at case 2 ('B') and falls through into case 3 ('C'), then hits break, so output is "BC".*

Q139 | Difficult & Tricky Questions | Tricky: Array Default

What is the output of the following code?

```
boolean[] flags = new boolean[3];
System.out.println(flags[1]);
```

A. null
B. 0
C. false
D. true

Ans: C

Explanation: *Boolean array elements default to false automatically when the array is created without explicit initialization.*

Q140 | Difficult & Tricky Questions | Tricky: Nested Ternary

What is the output of the following code?

```
int n = 0;
System.out.println(n > 0 ? "Positive" : n == 0 ? "Zero" : "Negative");
```

A. Zero
B. 0
C. Positive
D. Negative

Ans: A

Explanation: *n > 0 is false, so the nested ternary checks n == 0, which is true, returning "Zero".*

Q141 | Difficult & Tricky Questions | Tricky: Expression Evaluation

What is the output of the following code?

```
int a = 5;
int b = 2;
System.out.println(a / b + a % b);
```

A. 5
B. 2.5
C. 1
D. 3

Ans: D

Explanation: *a / b (integer division) = 2, and a % b = 1; adding gives 2 + 1 = 3.*

Q142 | Difficult & Tricky Questions | Tricky: Identifier Rule

Which of the following is a legal Java identifier, despite looking unusual?

A. $_value2
B. value#2
C. val ue2
D. 2$value

Ans: A

Explanation: *'$_value2' starts with a dollar sign (allowed) followed by underscore, letters, and digits — all legal identifier characters.*

Q143 | Difficult & Tricky Questions | Tricky: Array Bounds

What is the output of the following code?

```
int[] arr = {1, 2, 3};
int i = arr.length;
System.out.println(arr[i - 1]);
```

A. ArrayIndexOutOfBoundsException

B. Compilation error

C. 3

D. 0

Ans: C

Explanation: *arr.length is 3, so i-1 is 2, a valid index; arr[2] holds the value 3.*

Q144 | Difficult & Tricky Questions | Tricky: Logical Short-Circuit

What is the output of the following code?

```
int x = 5;
boolean result = (x > 10) && (x++ > 0);
System.out.println(x);
```

A. 6

B. 5

C. 10

D. 11

Ans: B

Explanation: *Since x > 10 is false, the && operator short-circuits and never evaluates x++, so x remains 5.*

Q145 | Difficult & Tricky Questions | Tricky: Data Type Range

What is most likely to happen when a value larger than int's range is assigned directly to an int variable as a literal?

A. The value is stored correctly without issue

B. Java automatically converts it to a long

C. The value silently wraps around at runtime

D. A compile-time error occurs because the literal is out of range

Ans: D

Explanation: *If an integer literal exceeds int's range and is not properly typed as long, the compiler flags it as an error.*

Q146 | Difficult & Tricky Questions | Tricky: Assertion-Reason

Assertion (A): The expression 5 / 2 evaluates to 2 in Java. Reason (R): Division between two integers performs integer division, discarding the remainder.

A. A is true but R is false

B. A is false but R is true

C. Both A and R are true, and R correctly explains A

D. Both A and R are false

Ans: C

Explanation: *5/2 does equal 2 in integer division, and this is precisely because Java truncates the fractional part in int/int division.*

Q147 | Difficult & Tricky Questions | Tricky: Coding Convention Edge Case

A programmer declares a variable as 'int Class1 = 10;'. Is this valid Java, and does it follow convention?

A. It is valid Java syntax, but it violates standard naming convention since it isn't camelCase

B. It is valid and fully follows convention

C. It is invalid because identifiers cannot contain digits

D. It is invalid because 'Class1' resembles the keyword 'class'

Ans: A

Explanation: *'Class1' is a legal identifier (not the reserved word 'class'), but starting with uppercase breaks Java's variable naming convention.*

Q148 | Difficult & Tricky Questions | Tricky: Array Length vs Index

What is the output of the following code?

```
int[] arr = new int[5];
System.out.println(arr.length - arr[0]);
```

A. 4

B. 5

C. Compilation error

D. 0

Ans: B

Explanation: *arr.length is 5 and arr[0] defaults to 0 (uninitialized int), so 5 - 0 = 5.*

Q149 | Difficult & Tricky Questions | Tricky: Multiple Conditions

What is the output of the following code?

```
int a = 10, b = 20, c = 30;
if (a < b && b < c || a > c)
System.out.println("True Path");
else
System.out.println("False Path");
```

A. 10

B. False Path

C. 30

D. True Path

Ans: D

Explanation: *a < b (true) && b < c (true) evaluates to true; since one side of || is already true, the overall condition is true.*

Section 9 — Mixed OAVS TGT Practice Set

Q150 | Mixed OAVS TGT Practice Set | Practice

Which of the following is a valid Java identifier commonly used for a loop counter?

A. in dex
B. i-1
C. i
D. 1i

Ans: C

Explanation: *A single letter like 'i' is a short but perfectly valid and common identifier for loop counters.*

Q151 | Mixed OAVS TGT Practice Set | Practice

Which primitive data type would best store a true/false flag indicating whether a student is present?

A. byte
B. boolean
C. int
D. char

Ans: B

Explanation: *A presence flag has exactly two states, making boolean the natural and most appropriate choice.*

Q152 | Mixed OAVS TGT Practice Set | Practice

According to Java coding conventions, how should the method name 'get student marks' be correctly written?

A. getStudentMarks
B. get_student_marks
C. GetStudentMarks
D. GETSTUDENTMARKS

Ans: A

Explanation: *Method names follow camelCase starting with a lowercase letter, so 'getStudentMarks' is correct.*

Q153 | Mixed OAVS TGT Practice Set | Practice

What is the output of the following code?

```
int a = 8, b = 3;
System.out.println(a - b * 2);
```

A. -2
B. 2
C. 13
D. 10

Ans: B

Explanation: *Multiplication happens first: b*2=6, then a-6 = 8-6 = 2.*

Q154 | Mixed OAVS TGT Practice Set | Practice

What is the output of the following code?

```
int marks = 55;
if (marks >= 90) System.out.println("A");
else if (marks >= 70) System.out.println("B");
else if (marks >= 50) System.out.println("C");
else System.out.println("D");
```

A. A

B. B

C. D

D. C

Ans: D

Explanation: *55 fails the first two checks but satisfies marks >= 50, so "C" is printed.*

Q155 | Mixed OAVS TGT Practice Set | Practice

What is the output of the following switch code?

```
char c = 'Z';
switch (c) {
case 'X': System.out.println(1); break;
case 'Y': System.out.println(2); break;
default: System.out.println(3);
}
```

A. Compilation error

B. 2

C. 3

D. 1

Ans: C

Explanation: *Since 'Z' matches neither 'X' nor 'Y', the default case executes and prints 3.*

Q156 | Mixed OAVS TGT Practice Set | Practice

What is the output of the following code?

```
int[] arr = {2, 4, 6};
int total = 0;
for (int i = 0; i < arr.length; i++)
total = total + arr[i];
System.out.println(total);
```

A. 12

B. 10

C. 6

D. 8

Ans: A

Explanation: *Summing all elements: 2 + 4 + 6 = 12.*

Q157 | Mixed OAVS TGT Practice Set | Practice

Which of the following statements about Java is correct regarding data type sizes?

A. long occupies more memory than int

B. short occupies more memory than int

C. byte occupies more memory than short

D. double occupies less memory than float

Ans: A

Explanation: *long (8 bytes) is larger than int (4 bytes); the other listed comparisons are all false.*

Q158 | Mixed OAVS TGT Practice Set | Practice

What is the output of the following code?

```
int x = 4;
x *= x + 1;
System.out.println(x);
```

A. 5

B. 16

C. 17

D. 20

Ans: D

Explanation: *x *= x + 1 means x = x * (x+1) = 4 * 5 = 20 (the right side is fully evaluated before multiplying).*

Q159 | Mixed OAVS TGT Practice Set | Practice

Which of the following is TRUE about array indices in Java?

A. Array indices start from 1 by default

B. Array indices can be negative

C. The last valid index of an array is (length - 1)

D. The last valid index equals the array length

Ans: C

Explanation: *Since indexing starts at 0, the final valid index of an array is always one less than its length.*